

✓ Lab Problem Statement 6 — Statistical Analysis of Palmer Penguins

Objective: Perform descriptive statistics, hypothesis testing, effect-size estimation, one-way ANOVA, Tukey HSD, Kruskal-Wallis, two-way ANOVA, and flipper-length analysis using the Palmer Penguins dataset..

✓ 1. Setup and Dataset

We use the `palmerpenguins` R package, which provides the Palmer Penguins dataset used in the lab statement.

```
# Install packages if needed
packages <- c("palmerpenguins", "dplyr", "ggplot2", "moments", "car", "effectsize")
new_packages <- packages[!(packages %in% installed.packages()[,"Package"])]
if(length(new_packages)) install.packages(new_packages, repos="https://cloud.r-project.org")

library(palmerpenguins)
library(dplyr)
library(ggplot2)
library(moments)
library(car)
library(effectsize)

data <- penguins

# Basic inspection
head(data)
dim(data)
str(data)
summary(data)
```

Installing packages into ‘/usr/local/lib/R/site-library’
(as ‘lib’ is unspecified)

also installing the dependencies ‘colorspace’, ‘fracdiff’, ‘lmtest’, ‘timeDate’, ‘urca’, ‘zoo’, ‘RcppArmadillo’, ‘cowplot’,

Attaching package: ‘palmerpenguins’

The following objects are masked from ‘package:datasets’:

penguins, penguins_raw

Attaching package: ‘dplyr’

The following objects are masked from ‘package:stats’:

filter, lag

The following objects are masked from ‘package:base’:

intersect, setdiff, setequal, union

Loading required package: carData

Attaching package: ‘car’

The following object is masked from ‘package:dplyr’:

recode

A tibble: 6 × 8

species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g	sex	year
<fct>	<fct>	<dbl>	<dbl>	<int>	<int>	<fct>	<int>
Adelie	Torgersen	39.1	18.7	181	3750	male	2007
Adelie	Torgersen	39.5	17.4	186	3800	female	2007
Adelie	Torgersen	40.3	18.0	195	3250	female	2007
Adelie	Torgersen	NA	NA	NA	NA	NA	2007
Adelie	Torgersen	36.7	19.3	193	3450	female	2007
Adelie	Torgersen	39.3	20.6	190	3650	male	2007

344 × 8

tibble [344 × 8] (S3: tbl_df/tbl/data.frame)

```
$ species      : Factor w/ 3 levels "Adelie","Chinstrap",...: 1 1 1 1 1 1 1 1 1 1 ...
$ island       : Factor w/ 3 levels "Biscoe","Dream",...: 3 3 3 3 3 3 3 3 3 3 ...
$ bill_length_mm : num [1:344] 39.1 39.5 40.3 NA 36.7 39.3 38.9 39.2 34.1 42 ...
$ bill_depth_mm  : num [1:344] 18.7 17.4 18 NA 19.3 20.6 17.8 19.6 18.1 20.2 ...
$ flipper_length_mm: int [1:344] 181 186 195 NA 193 190 181 195 193 190 ...
$ body_mass_g    : int [1:344] 3750 3800 3250 NA 3450 3650 3625 4675 3475 4250 ...
$ sex           : Factor w/ 2 levels "female","male": 2 1 1 NA 1 2 1 2 NA NA ...
$ year          : int [1:344] 2007 2007 2007 2007 2007 2007 2007 2007 2007 2007 ...
```

	species	island	bill_length_mm	bill_depth_mm
Adelie	:152	Biscoe	:168	Min. :32.10
Chinstrap	:68	Dream	:124	1st Qu.:39.23
Gentoo	:124	Torgersen	:52	Median :44.45
				Mean :43.92
				3rd Qu.:48.50
				Max. :59.60
				NAs :2

	flipper_length_mm	body_mass_g	sex	year
Min.	:172.0	Min. :2700	female:165	Min. :2007
1st Qu.	:190.0	1st Qu.:3550	male :168	1st Qu.:2007
Median	:197.0	Median :4050	NAs : 11	Median :2008
Mean	:200.9	Mean :4202		Mean :2008
3rd Qu.	:213.0	3rd Qu.:4750		3rd Qu.:2009
Max.	:231.0	Max. :6300		Max. :2009
NAs	:2	NAs :2		

2. Data Cleaning

For statistical tests involving body mass, sex, or flipper length, rows with missing values in the required variables are removed.

```
penguins_clean <- data %>%
  filter(!is.na(body_mass_g),
         !is.na(flipper_length_mm),
         !is.na(species),
         !is.na(sex))

cat("Original rows:", nrow(data), "\n")
cat("Complete rows for main analysis:", nrow(penguins_clean), "\n")

table(penguins_clean$species)
table(penguins_clean$sex)
```

Original rows: 344

Complete rows for main analysis: 333

```
      Adelie Chinstrap   Gentoo
      146         68      119

female   male
      165    168
```

3. Descriptive Statistical Analysis of Body Mass

Required measures: mean, median, minimum, maximum, variance, standard deviation, quartiles, IQR, skewness, and kurtosis.

```
# Overall descriptive statistics
body_mass_stats <- data %>%
  filter(!is.na(body_mass_g)) %>%
  summarise(
    Mean = mean(body_mass_g),
    Median = median(body_mass_g),
    Minimum = min(body_mass_g),
    Maximum = max(body_mass_g),
    Variance = var(body_mass_g),
    SD = sd(body_mass_g),
    Q1 = quantile(body_mass_g, 0.25),
    Q3 = quantile(body_mass_g, 0.75),
    IQR = IQR(body_mass_g),
    Skewness = skewness(body_mass_g),
    Kurtosis = kurtosis(body_mass_g)
  )

body_mass_stats
```

A tibble: 1 × 11

Mean	Median	Minimum	Maximum	Variance	SD	Q1	Q3	IQR	Skewness	Kurtosis
<dbl>	<dbl>	<int>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
4201.754	4050	2700	6300	643131.1	801.9545	3550	4750	1200	0.468264	2.273757

```
# Species-wise descriptive statistics
species_stats <- data %>%
  filter(!is.na(body_mass_g)) %>%
  group_by(species) %>%
  summarise(
    N = n(),
    Mean = mean(body_mass_g),
    Median = median(body_mass_g),
    Minimum = min(body_mass_g),
    Maximum = max(body_mass_g),
    Variance = var(body_mass_g),
    SD = sd(body_mass_g),
    Q1 = quantile(body_mass_g, 0.25),
    Q3 = quantile(body_mass_g, 0.75),
    IQR = IQR(body_mass_g),
    Skewness = skewness(body_mass_g),
    Kurtosis = kurtosis(body_mass_g),
    .groups = "drop"
  )
```

species_stats

A tibble: 3 × 13

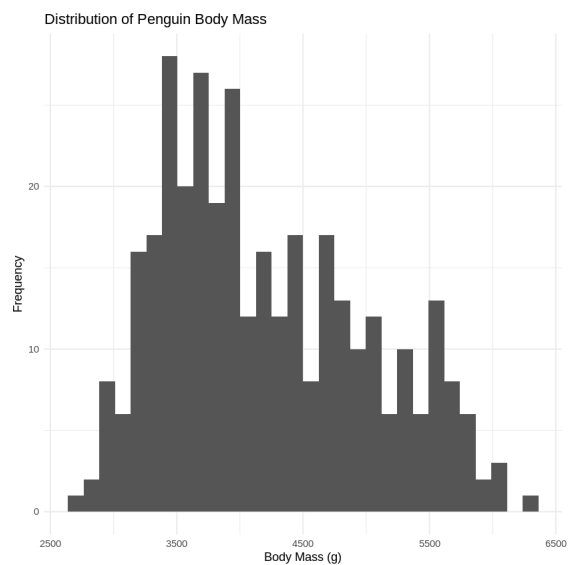
species	N	Mean	Median	Minimum	Maximum	Variance	SD	Q1	Q3	IQR	Skewness	Kurtosis
<fct>	<int>	<dbl>	<dbl>	<int>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
Adelie	151	3700.662	3700	2850	4775	210282.9	458.5661	3350.0	4000	650.0	0.28249381	2.405611
Chinstrap	68	3733.088	3700	2700	4800	147713.5	384.3351	3487.5	3950	462.5	0.24194125	3.463681
Gentoo	123	5076.016	5000	3950	6300	254133.2	504.1162	4700.0	5500	800.0	0.06878276	2.257871

Interpretation

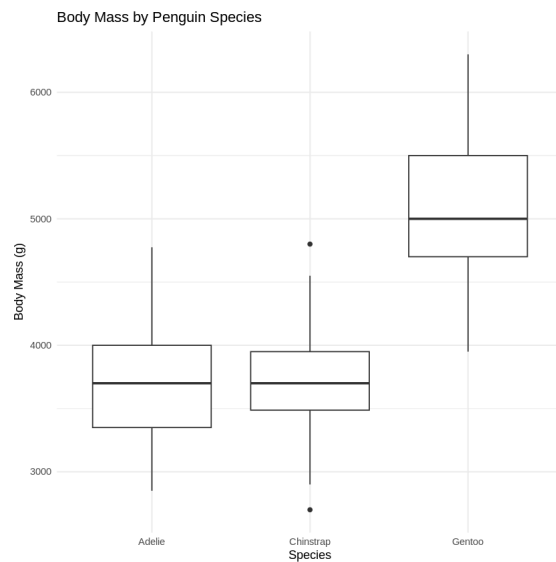
The table above provides the central tendency and dispersion of penguin body mass overall and separately by species. Comparing the species means, medians, and boxplot distributions helps identify whether some species are systematically heavier.

4. Visualizations — Body Mass

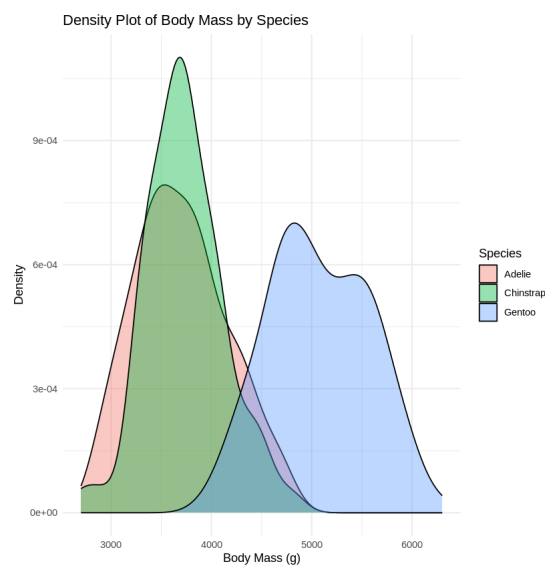
```
# Histogram
ggplot(data, aes(x = body_mass_g)) +
  geom_histogram(bins = 30, na.rm = TRUE) +
  labs(
    title = "Distribution of Penguin Body Mass",
    x = "Body Mass (g)",
    y = "Frequency"
  ) +
  theme_minimal()
```



```
# Species-wise boxplot
ggplot(data, aes(x = species, y = body_mass_g)) +
  geom_boxplot(na.rm = TRUE) +
  labs(
    title = "Body Mass by Penguin Species",
    x = "Species",
    y = "Body Mass (g)"
  ) +
  theme_minimal()
```



```
# Density plot
ggplot(data, aes(x = body_mass_g, fill = species)) +
  geom_density(alpha = 0.4, na.rm = TRUE) +
  labs(
    title = "Density Plot of Body Mass by Species",
    x = "Body Mass (g)",
    y = "Density",
    fill = "Species"
  ) +
  theme_minimal()
```



5. Hypothesis Testing — Male vs Female Body Mass

H₀: Mean body mass of male penguins equals mean body mass of female penguins.

H₁: Mean body mass of male penguins differs from mean body mass of female penguins.

We first inspect normality using Shapiro-Wilk tests and QQ-plots, then perform an independent two-sample Welch t-test.

```
sex_data <- data %>%
  filter(!is.na(body_mass_g), !is.na(sex))

# Shapiro-Wilk tests
shapiro_male <- shapiro.test(sex_data$body_mass_g[sex_data$sex == "male"])
shapiro_female <- shapiro.test(sex_data$body_mass_g[sex_data$sex == "female"])

shapiro_male
shapiro_female
```

Shapiro-Wilk normality test

```
data: sex_data$body_mass_g[sex_data$sex == "male"]
W = 0.92504, p-value = 1.227e-07
```

Shapiro-Wilk normality test

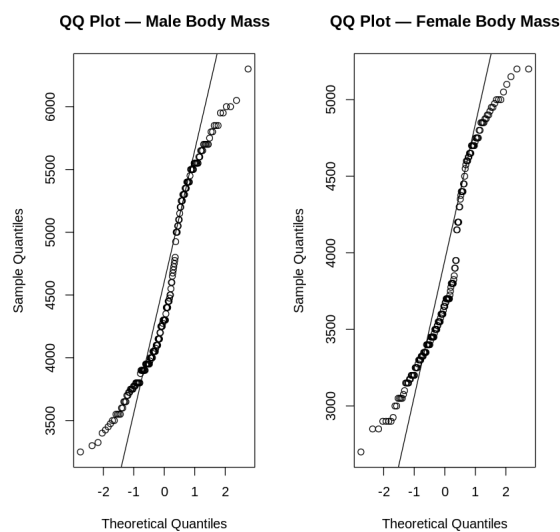
```
data: sex_data$body_mass_g[sex_data$sex == "female"]
W = 0.91931, p-value = 6.155e-08
```

```
# QQ-plots
par(mfrow = c(1, 2))

qqnorm(sex_data$body_mass_g[sex_data$sex == "male"],
       main = "QQ Plot - Male Body Mass")
qqline(sex_data$body_mass_g[sex_data$sex == "male"])

qqnorm(sex_data$body_mass_g[sex_data$sex == "female"],
       main = "QQ Plot - Female Body Mass")
qqline(sex_data$body_mass_g[sex_data$sex == "female"])

par(mfrow = c(1, 1))
```



```
# Welch independent two-sample t-test
t_test <- t.test(
  body_mass_g ~ sex,
  data = sex_data,
  var.equal = FALSE,
  conf.level = 0.95
)

t_test
```

Welch Two Sample t-test

```
data: body_mass_g by sex
t = -8.5545, df = 323.9, p-value = 4.794e-16
alternative hypothesis: true difference in means between group female and group male is not equal to 0
95 percent confidence interval:
 -840.5783 -526.2453
sample estimates:
mean in group female    mean in group male
      3862.273           4545.685
```

```
# Cohen's d
cohen_d <- cohens_d(
  body_mass_g ~ sex,
  data = sex_data,
  pooled_sd = TRUE
)

cohen_d
```

A effectsize_difference: 1 × 4

Cohens_d	CI	CI_low	CI_high
<dbl>	<dbl>	<dbl>	<dbl>
-0.9362048	0.95	-1.161904	-0.709225

Interpretation of the t-test

- If the p-value is below 0.05, reject H_0 and conclude that male and female mean body masses differ significantly.
- The 95% confidence interval gives the plausible range for the difference in population means.
- Cohen's d describes the magnitude of the difference: approximately 0.2 = small, 0.5 = medium, and 0.8 = large effect as a common rule of thumb.

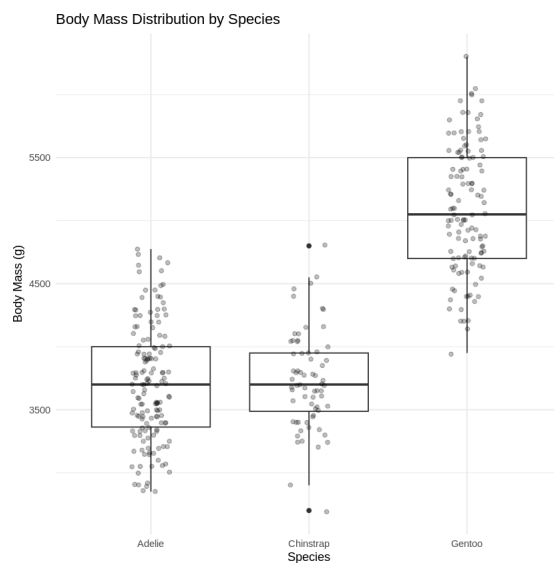
6. One-Way ANOVA — Body Mass Across Species

Model:

```
body_mass_g ~ species
```

Before ANOVA, we examine distributions, normality, and homogeneity of variance.

```
# Distribution by species
ggplot(penguins_clean, aes(x = species, y = body_mass_g)) +
  geom_boxplot() +
  geom_jitter(width = 0.12, alpha = 0.25) +
  labs(
    title = "Body Mass Distribution by Species",
    x = "Species",
    y = "Body Mass (g)"
  ) +
  theme_minimal()
```



```
# Shapiro-Wilk normality test within each species
by(penguins_clean$body_mass_g,
    penguins_clean$species,
    shapiro.test)
```

```
penguins_clean$species: Adelie
```

```
Shapiro-Wilk normality test
```

```
data: dd[x, ]
W = 0.98116, p-value = 0.04232
```

```
-----
penguins_clean$species: Chinstrap
```

```
Shapiro-Wilk normality test
```

```
data: dd[x, ]
W = 0.98449, p-value = 0.5605
```

```
-----
penguins_clean$species: Gentoo
```

```
Shapiro-Wilk normality test
```

```
data: dd[x, ]
W = 0.98606, p-value = 0.2605
```

```
# Levene's test for homogeneity of variance
levene_result <- leveneTest(body_mass_g ~ species, data = penguins_clean)
levene_result
```

```
Aanova: 2 × 3
```

	Df	F value	Pr(>F)
	<int>	<dbl>	<dbl>
group	2	5.134899	0.006367054
	330	NA	NA

```
# One-way ANOVA
anova_one <- aov(body_mass_g ~ species, data = penguins_clean)
summary(anova_one)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
species	2	145190219	72595110	341.9	<2e-16 ***
Residuals	330	70069447	212332		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

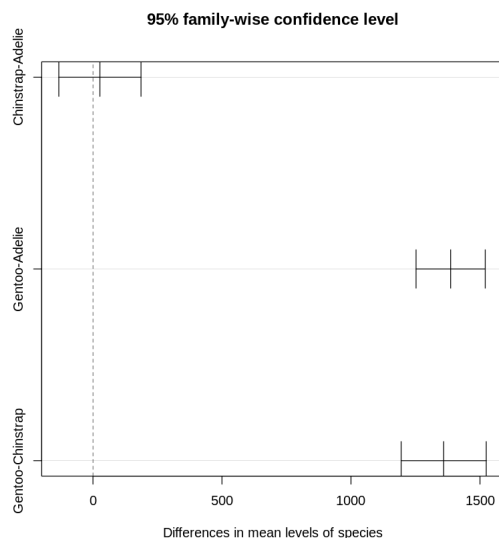
```
# Tukey HSD post-hoc comparison
tukey_result <- TukeyHSD(anova_one)
tukey_result
plot(tukey_result)
```


Tukey multiple comparisons of means
95% family-wise confidence level

```
Fit: aov(formula = body_mass_g ~ species, data = penguins_clean)
```

```
$species
```

	diff	lwr	upr	p adj
Chinstrap-Adelie	26.92385	-132.3528	186.2005	0.916431
Gentoo-Adelie	1386.27259	1252.2897	1520.2554	0.000000
Gentoo-Chinstrap	1359.34874	1194.4304	1524.2671	0.000000



One-Way ANOVA Interpretation

The ANOVA tests whether all three species have the same population mean body mass. If the ANOVA p-value is below 0.05, the null hypothesis is rejected and at least one species mean differs.

Tukey HSD identifies the specific species pairs responsible for the significant overall result.

7. Non-Parametric Analysis — Kruskal-Wallis

If ANOVA assumptions are substantially violated, the Kruskal-Wallis test provides a non-parametric alternative for comparing the distributions across the three species.

```
kruskal_result <- kruskal.test(body_mass_g ~ species, data = penguins_clean)
kruskal_result
```

Kruskal-Wallis rank sum test

```
data: body_mass_g by species
Kruskal-Wallis chi-squared = 212.09, df = 2, p-value < 2.2e-16
```

```
# Side-by-side summary of p-values
anova_p <- summary(anova_one)[[1]][["Pr(>F)"]][1]
kruskal_p <- kruskal_result$p.value

cat("One-way ANOVA p-value:", anova_p, "\n")
cat("Kruskal-Wallis p-value:", kruskal_p, "\n")
```

```
One-way ANOVA p-value: 3.744505e-81
Kruskal-Wallis p-value: 8.836877e-47
```

Comparison

If both p-values are below 0.05, both parametric and non-parametric analyses support the conclusion that body mass differs among species. If their conclusions differ, the assumptions and the nature of the distributions should be examined more carefully.

8. Two-Way ANOVA — Species and Sex

Model:

```
body_mass_g ~ species * sex
```

The interaction term tests whether the effect of sex on body mass depends on penguin species.

```
two_way <- aov(body_mass_g ~ species * sex, data = penguins_clean)
summary(two_way)
```

```

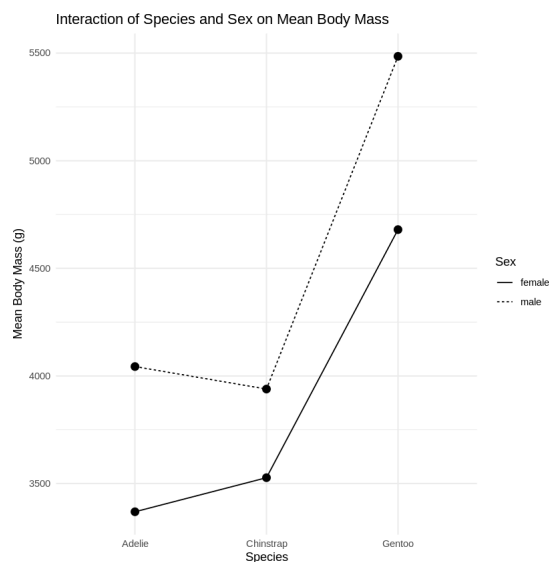
      Df    Sum Sq Mean Sq F value    Pr(>F)
species  2 145190219  72595110  758.358 < 2e-16 ***
sex       1  37090262  37090262  387.460 < 2e-16 ***
species:sex  2   1676557    838278    8.757 0.000197 ***
Residuals 327  31302628    95727
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

# Interaction visualization
ggplot(penguins_clean,
       aes(x = species, y = body_mass_g, group = sex, linetype = sex)) +
  stat_summary(fun = mean, geom = "line") +
  stat_summary(fun = mean, geom = "point", size = 3) +
  labs(
    title = "Interaction of Species and Sex on Mean Body Mass",
    x = "Species",
    y = "Mean Body Mass (g)",
    linetype = "Sex"
  ) +
  theme_minimal()

```



Two-Way ANOVA Interpretation

Report the p-values for:

1. **Species main effect** — whether mean body mass differs by species.
2. **Sex main effect** — whether mean body mass differs by sex.
3. **Species × Sex interaction** — whether the sex difference changes across species.

A significant interaction should be interpreted before relying on the main effects alone.

9. Additional Analysis — Flipper Length

We repeat the species comparison using `flipper_length_mm`.

```

flipper_data <- data %>%
  filter(!is.na(flipper_length_mm), !is.na(species))

# Descriptive statistics
flipper_stats <- flipper_data %>%
  group_by(species) %>%
  summarise(
    N = n(),

```

```

    Mean = mean(flipper_length_mm),
    Median = median(flipper_length_mm),
    SD = sd(flipper_length_mm),
    Minimum = min(flipper_length_mm),
    Maximum = max(flipper_length_mm),
    IQR = IQR(flipper_length_mm),
    .groups = "drop"
  )
}

flipper_stats

```

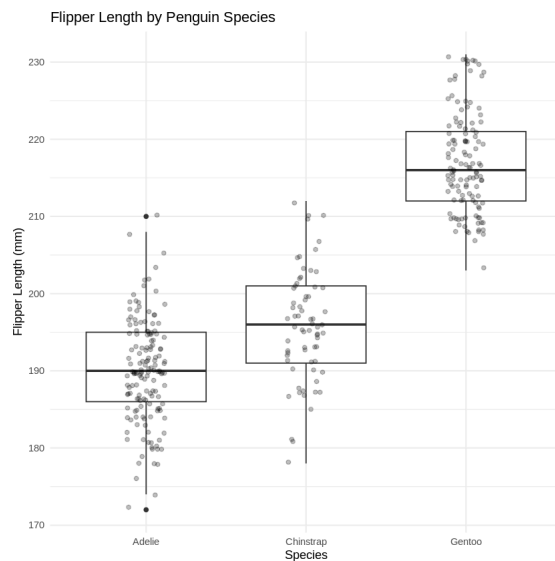
A tibble: 3 × 8

species	N	Mean	Median	SD	Minimum	Maximum	IQR
<fct>	<int>	<dbl>	<dbl>	<dbl>	<int>	<int>	<dbl>
Adelie	151	189.9536	190	6.539457	172	210	9
Chinstrap	68	195.8235	196	7.131894	178	212	10
Gentoo	123	217.1870	216	6.484976	203	231	9

```

# Species-wise flipper length comparison
ggplot(flipper_data, aes(x = species, y = flipper_length_mm)) +
  geom_boxplot() +
  geom_jitter(width = 0.12, alpha = 0.25) +
  labs(
    title = "Flipper Length by Penguin Species",
    x = "Species",
    y = "Flipper Length (mm)"
  ) +
  theme_minimal()

```



```

# One-way ANOVA for flipper length
flipper_anova <- aov(flipper_length_mm ~ species, data = flipper_data)
summary(flipper_anova)

```

```

# Tukey HSD
TukeyHSD(flipper_anova)

```

```

      Df Sum Sq Mean Sq F value Pr(>F)
species    2  52473    26237   594.8 <2e-16 ***
Residuals 339  14953         44
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Tukey multiple comparisons of means
 95% family-wise confidence level

Fit: aov(formula = flipper_length_mm ~ species, data = flipper_data)

$species
      diff      lwr      upr p adj
Chinstrap-Adelie  5.869887  3.586583  8.153191    0
Gentoo-Adelie    27.233349  25.334376  29.132323    0
Gentoo-Chinstrap 21.363462  19.000841  23.726084    0

```

10. Group Comparison Plot with Means and 95% CI

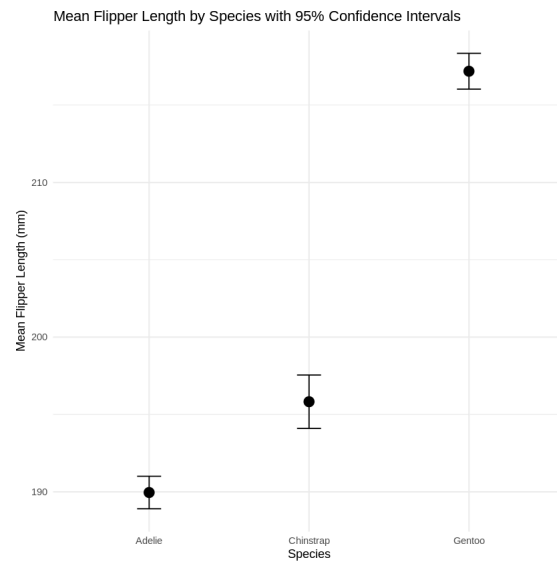
```
# Calculate mean and 95% CI by species
flipper_ci <- flipper_data %>%
  group_by(species) %>%
  summarise(
    n = n(),
    mean = mean(flipper_length_mm),
    sd = sd(flipper_length_mm),
    se = sd / sqrt(n),
    lower = mean - qt(0.975, df = n - 1) * se,
    upper = mean + qt(0.975, df = n - 1) * se,
    .groups = "drop"
  )

flipper_ci
```

A tibble: 3 × 7

species	n	mean	sd	se	lower	upper
<fct>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
Adelie	151	189.9536	6.539457	0.5321735	188.9021	191.0052
Chinstrap	68	195.8235	7.131894	0.8648692	194.0972	197.5498
Gentoo	123	217.1870	6.484976	0.5847306	216.0295	218.3445

```
ggplot(flipper_ci, aes(x = species, y = mean)) +
  geom_point(size = 4) +
  geom_errorbar(aes(ymin = lower, ymax = upper), width = 0.15) +
  labs(
    title = "Mean Flipper Length by Species with 95% Confidence Intervals",
    x = "Species",
    y = "Mean Flipper Length (mm)"
  ) +
  theme_minimal()
```



11. Statistical Summary

```
cat("==== FINAL STATISTICAL SUMMARY ====\n\n")
```